

FIRST BREADBOARD PROTOTYPE - 3V3 DK USB POWER

VERIFY: DCF OUT LOGIC LEVEL BEFORE MCU CONNECTION. PON IS ACTIVE-LOW. VERIFY HAT BS1=0 FOR 4-WIRE SPI.

POWER: USE DK 3V3 FOR INTEGRATED TESTS. DO NOT CONNECT KORAD AND DK 3V3 TOGETHER. BUZZER RATING AND Q1 PINOUT MUST BE VERIFIED.

The schematic shows the following components and connections:

- J1: nRF52 DK PCA10040 interface**
 - 15 GND
 - 14 BUZZER_EN
 - 13 UNVERIFIED_LEVEL
 - 12 PON_ACTIVE_LOW
 - 11 SENSOR_INT1
 - 10 SENSOR_CS
 - 9 EPD_BUSY
 - 8 EPD_RST
 - 7 EPD_DC
 - 6 EPD_CS
 - 5 SPI_MISO
 - 4 SPI_MOSI
 - 3 SPI_SCK
 - 2 GND
 - 1 3V3
- J2: Waveshare 2.13inch HAT Rev2.1 (BS1=0)**
 - 8 EPD_BUSY
 - 7 EPD_RST
 - 6 EPD_DC
 - 5 EPD_CS
 - 4 CLK
 - 3 SPI_SCK
 - 2 GND
 - 1 3V3
- J3: Adafruit LIS3DH breakout**
 - 9 INT2
 - 8 INT1
 - 7 CS
 - 6 SDO
 - 5 SDA_SDI
 - 4 SCL_SCK
 - 3 GND
 - 2 3Vo
 - 1 VIN
- J4: DCF-1060N-800 (PON active-low)**
 - 4 VDD_1V1_3V3
 - 3 GND
 - 2 OUT
 - 1 PON_ACTIVE_LOW
- R1: 1k BASE**
 - 1 BUZZER_EN
 - 2 BUZZER_BASE
- R2: 10k PON PULL-UP OPTIONAL**
 - 1 3V3
 - 2 DCF_OUT_UNVERIFIED_LEVEL
- R3: 100k BASE PULL-DOWN OPTIONAL**
 - 1 BUZZER_BASE
 - 2 GND
- D1: AP-1205V-P1 VERIFY RATING**
 - 1 3V3
 - 2 BUZZER_SWITCHED
- BZ1: 1N4148 FLYBACK**
 - 1 3V3
 - 2 BUZZER_SWITCHED
- Q1: 2N3904**
 - 1 BUZZER_SWITCHED
 - 2 BUZZER_BASE

The schematic is titled "FIRST BREADBOARD PROTOTYPE - 3V3 DK USB POWER" and includes a note about verifying DCF OUT logic level before MCU connection. The schematic is color-coded: yellow for components, green for connections, and red for power and ground connections.

FIRST BREADBOARD PROTOTYPE – 3V3 DK USB POWER

VERIFY: DCF OUT LOGIC LEVEL BEFORE MCU CONNECTION. PON IS ACTIVE-LOW. VERIFY HAT BS1=0 FOR 4-WIRE SPI.

POWER: USE DK 3V3 FOR INTEGRATED TESTS. DO NOT CONNECT KORAD AND DK 3V3 TOGETHER. BUZZER RATING AND Q1 PINOUT MUST BE VERIFIED.

The schematic shows the following components and connections:

- J1: nRF52 DK PCA10040 interface**
 - 15 GND
 - 14 BUZZER_EN
 - 13 UNVERIFIED_LEVEL
 - 12 PON_ACTIVE_LOW
 - 11 SENSOR_INT1
 - 10 SENSOR_CS
 - 9 EPD_BUSY
 - 8 EPD_RST
 - 7 EPD_DC
 - 6 EPD_CS
 - 5 EPD_CS
 - 4 SENSOR_MISO
 - 3 EPD_MOSI
 - 2 EPD_SCK
 - 1 GND
 - 3V3
- J2: WaveShare 2.13inch HAT Rev2.1 (BS1=0)**
 - 8 EPD_BUSY
 - 7 EPD_RST
 - 6 EPD_DC
 - 5 EPD_CS
 - 4 CS
 - 3 CLK
 - 2 DIN
 - 1 GND
 - 3V3
- J3: Adafruit LIS3DH breakout**
 - 9 INT2
 - 8 INT1
 - 7 CS
 - 6 SDO
 - 5 SDA_SDI
 - 4 SCL_SCK
 - 3 GND
 - 2 3Vo
 - 1 VIN
- J4: DCF-1060N-800 (PON active-low)**
 - 4 3V3
 - 3 GND
 - 2 DCF_OUT_UNVERIFIED_LEVEL
 - 1 DCF_PON_ACTIVE_LOW
- R1: 1k BASE**
 - 1 BUZZER_EN
 - 2 BUZZER_BASE
- R2: 10k PON PULL-UP OPTIONAL**
 - 1 3V3
 - 2 DCF_PON_ACTIVE_LOW
- R3: 100k BASE PULL-DOWN OPTIONAL**
 - 1 BUZZER_BASE
 - 2 GND
- BZ1: Buzzer**
 - 2 BUZZER_SWITCHED
 - 1 3V3
- D1: AP-1205V-P1 VERIFY RATING**
 - 2 BUZZER_SWITCHED
 - 1 3V3
- Q1: 2N3904**
 - 1 BUZZER_SWITCHED
 - 2 BUZZER_BASE

Sheet: /
File: breadboard-prototype.kicad_sch
Title:
Size: A4 **Date:** **Rev:**
KiCad E.D.A. 10.0.4 **Id: 1/1**

FIRST BREADBOARD PROTOTYPE - 3V3 DK USB POWER

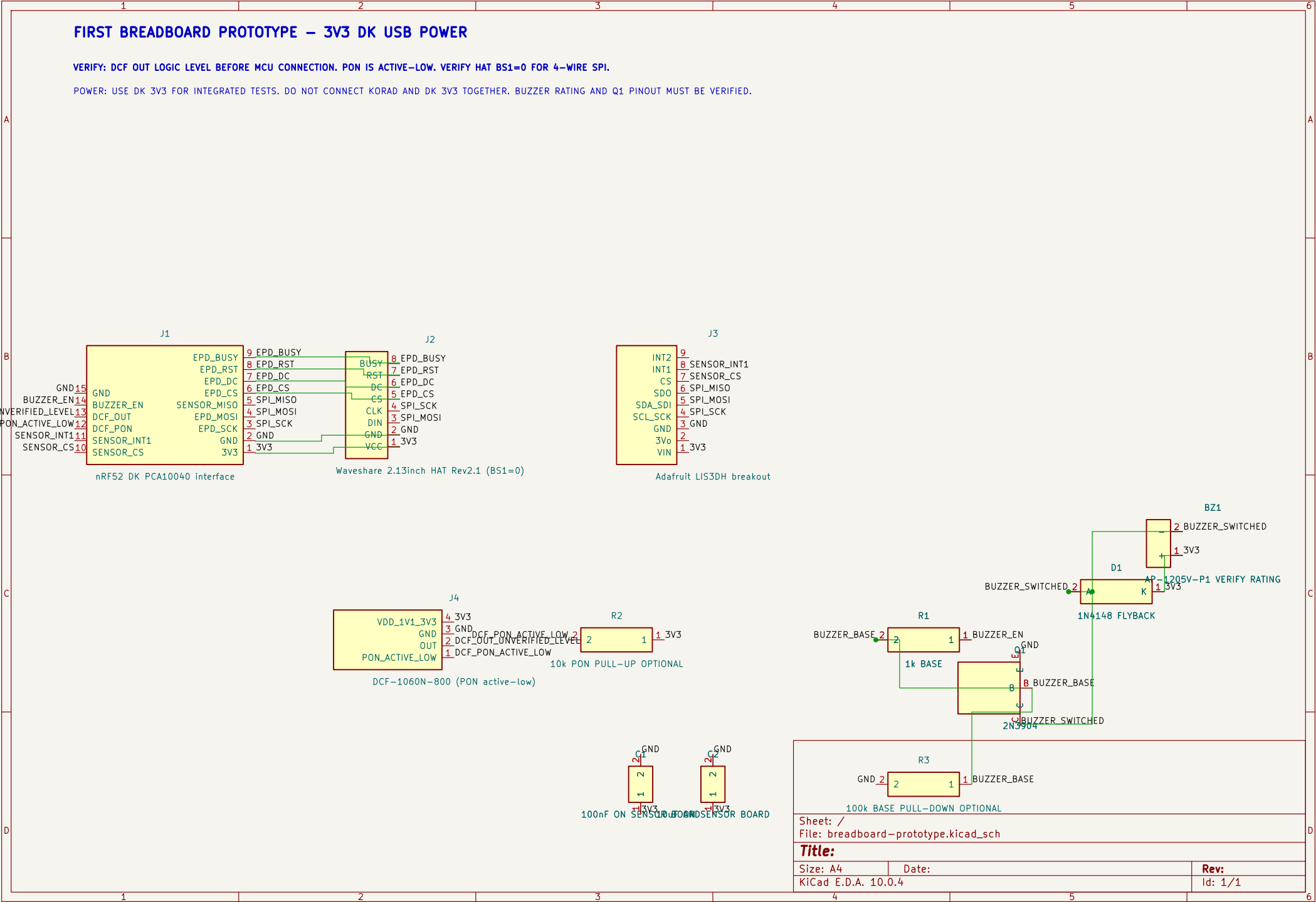
VERIFY: DCF OUT LOGIC LEVEL BEFORE MCU CONNECTION. PON IS ACTIVE-LOW. VERIFY HAT BS1=0 FOR 4-WIRE SPI.

POWER: USE DK 3V3 FOR INTEGRATED TESTS. DO NOT CONNECT KORAD AND DK 3V3 TOGETHER. BUZZER RATING AND Q1 PINOUT MUST BE VERIFIED.

Components and Connections:

- J1: nRF52 DK PCA10040 interface**
 - 15 GND
 - 14 BUZZER_EN
 - 13 UNVERIFIED_LEVEL
 - 12 PON_ACTIVE_LOW
 - 11 SENSOR_INT1
 - 10 SENSOR_CS
 - 9 EPD_BUSY
 - 8 EPD_RST
 - 7 EPD_DC
 - 6 EPD_CS
 - 5 SPI_MISO
 - 4 SPI_MOSI
 - 3 SPI_SCK
 - 2 GND
 - 1 3V3
- J2: WaveShare 2.13inch HAT Rev2.1 (BS1=0)**
 - 8 EPD_BUSY
 - 7 EPD_RST
 - 6 EPD_DC
 - 5 EPD_CS
 - 4 CLK
 - 3 SPI_SCK
 - 2 GND
 - 1 3V3
- J3: Adafruit LIS3DH breakout**
 - 9 INT2
 - 8 INT1
 - 7 CS
 - 6 SDO
 - 5 SDA_SDI
 - 4 SCL_SCK
 - 3 GND
 - 2 3Vo
 - 1 3V3
- J4: DCF-1060N-800 (PON active-low)**
 - 4 3V3
 - 3 GND
 - 2 DCF_OUT_UNVERIFIED_LEVEL
 - 1 DCF_PON_ACTIVE_LOW
- R1: 1k BASE**
 - 1 BUZZER_EN
 - 2 BUZZER_BASE
- R2: 10k PON PULL-UP OPTIONAL**
 - 1 3V3
 - 2 DCF_PON_ACTIVE_LOW
- R3: 100k BASE PULL-DOWN OPTIONAL**
 - 1 BUZZER_BASE
 - 2 GND
- BZ1: Buzzer**
 - 2 BUZZER_SWITCHED
 - 1 3V3
- D1: AP-1205V-P1 VERIFY RATING**
 - 2 BUZZER_SWITCHED
 - 1 3V3
- C1, C2: 100nF ON SENSOR BOARD**
 - 1 3V3
 - 2 GND

Sheet: /
File: breadboard-prototype.kicad_sch
Title:
Size: A4 **Date:** **Rev:**
KiCad E.D.A. 10.0.4 **Id: 1/1**



FIRST BREADBOARD PROTOTYPE - 3V3 DK USB POWER

VERIFY: DCF OUT LOGIC LEVEL BEFORE MCU CONNECTION. PON IS ACTIVE-LOW. VERIFY HAT BS1=0 FOR 4-WIRE SPI.

POWER: USE DK 3V3 FOR INTEGRATED TESTS. DO NOT CONNECT KORAD AND DK 3V3 TOGETHER. BUZZER RATING AND Q1 PINOUT MUST BE VERIFIED.

Components and Connections:

- J1: nRF52 DK PCA10040 interface**
 - 15 GND
 - 14 BUZZER_EN
 - 13 UNVERIFIED_LEVEL
 - 12 PON_ACTIVE_LOW
 - 11 SENSOR_INT1
 - 10 SENSOR_CS
 - 9 EPD_BUSY
 - 8 EPD_RST
 - 7 EPD_DC
 - 6 EPD_CS
 - 5 SPI_MISO
 - 4 SPI_MOSI
 - 3 SPI_SCK
 - 2 GND
 - 1 3V3
- J2: WaveShare 2.13inch HAT Rev2.1 (BS1=0)**
 - 8 EPD_BUSY
 - 7 EPD_RST
 - 6 EPD_DC
 - 5 EPD_CS
 - 4 CLK
 - 3 SPI_SCK
 - 2 GND
 - 1 3V3
- J3: Adafruit LIS3DH breakout**
 - 9 INT2
 - 8 INT1
 - 7 CS
 - 6 SDO
 - 5 SDA_SDI
 - 4 SCL_SCK
 - 3 GND
 - 2 3Vo
 - 1 3V3
- J4: DCF-1060N-800 (PON active-low)**
 - 4 3V3
 - 3 GND
 - 2 DCF_OUT_UNVERIFIED_LEVEL
 - 1 DCF_PON_ACTIVE_LOW
- R1: 1k BASE**
 - 1 BUZZER_EN
 - 2 BUZZER_BASE
- R2: 10k PON PULL-UP OPTIONAL**
 - 1 3V3
 - 2 DCF_PON_ACTIVE_LOW
- R3: 100k BASE PULL-DOWN OPTIONAL**
 - 1 BUZZER_BASE
 - 2 GND
- BZ1: Buzzer**
 - 2 BUZZER_SWITCHED
 - 1 3V3
- D1: AP-1205V-P1 VERIFY RATING**
 - 2 BUZZER_SWITCHED
 - 1 3V3
- C1, C2: 100nF ON SENSOR BOARD**
 - 1 3V3
 - 2 GND

Sheet: /
File: breadboard-prototype.kicad_sch
Title:
Size: A4 **Date:** **Rev:**
KiCad E.D.A. 10.0.4 **Id: 1/1**